

HAMMERFEST, Norway

WMO: 010520

Lat: 70.683N

Lon: 23.683E

Elev: 81

StdP: 100.36

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.2	-11.8	-18.9	0.7	-12.2	-16.5	0.9	-11.0	17.1	1.4	15.1	0.1	3.5	110	0.793

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.6	21.0	14.7	19.1	13.8	17.1	12.7	15.3	19.9	14.1	18.4	12.9	16.6	3.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.0	9.4	18.1	11.8	8.7	16.1	10.8	8.1	14.7	43.3	19.9	39.8	18.7	36.7	16.6	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.6	11.6	10.1	-16.2	24.3	1.8	3.0	-17.5	26.5	-18.5	28.2	-19.5	29.9	-20.8	32.1		
(5)			-17.1	17.4	1.7	2.2	-18.3	19.0	-19.3	20.2	-20.2	21.4	-21.4	23.0	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-3.9	-4.3	-2.8	-0.1	4.0	7.9	11.4	10.6	7.9	3.1	-0.8	-1.9
	DBStd	6.47	4.00	4.10	3.59	3.15	3.46	3.29	3.66	3.26	3.10	3.39	3.20	3.53
	HDD10.0	2848	430	402	397	302	191	83	24	32	78	214	324	371
	HDD18.3	5738	688	635	656	552	445	314	218	242	314	472	574	629
	CDD10.0	154	0	0	0	0	3	18	68	49	14	1	0	0
	CDD18.3	4	0	0	0	0	0	1	3	1	0	0	0	0
	CDH23.3	21	0	0	0	0	0	1	17	3	0	0	0	0
	CDH26.7	2	0	0	0	0	0	0	1	1	0	0	0	0
Wind	WSAvg	4.5	4.8	4.9	5.2	4.4	4.4	4.3	3.9	3.8	4.0	4.6	4.4	4.9
Precipitation	PrecAvg	712	69	56	63	49	42	45	51	54	64	82	65	73
	PrecMax	828	141	96	123	90	99	104	136	133	111	133	120	115
	PrecMin	586	2	9	14	14	10	17	4	25	29	33	8	13
	PrecStd	68	38	21	27	20	20	21	29	24	21	22	33	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.8	5.2	6.1	10.0	17.3	21.9	25.7	22.2	18.0	12.8	7.9	6.3
		MCWB	4.0	3.1	3.3	5.7	10.8	14.1	18.3	14.9	13.5	9.4	6.2	4.5
	2%	DB	4.1	3.9	4.7	7.2	13.7	18.2	21.9	19.8	15.8	10.8	6.2	5.2
		MCWB	2.5	2.4	2.6	4.4	8.8	12.3	15.8	14.3	11.8	8.0	4.7	3.4
	5%	DB	3.0	2.7	3.2	5.3	11.1	15.6	20.0	17.9	14.1	9.0	4.9	4.0
		MCWB	1.6	1.3	1.4	3.2	7.2	10.9	14.5	13.2	10.6	7.1	3.4	2.4
	10%	DB	1.9	1.2	2.1	4.1	9.0	13.2	18.1	15.9	12.5	7.9	3.7	2.9
		MCWB	0.4	-0.2	0.4	2.3	6.0	9.5	13.5	12.2	9.7	6.4	2.3	1.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	3.7	3.7	6.2	11.1	15.2	18.7	16.2	13.6	9.5	6.4	4.9
		MCDB	5.2	4.6	5.1	9.3	15.9	20.6	24.1	20.3	17.0	12.2	7.4	6.2
	2%	WB	2.8	2.4	2.8	4.7	9.0	12.8	16.2	14.6	12.1	8.3	4.8	3.6
		MCDB	3.8	3.5	4.4	6.8	13.3	17.4	21.1	18.9	15.1	10.2	6.0	4.9
	5%	WB	1.6	1.3	1.6	3.4	7.5	11.1	14.8	13.4	11.0	7.2	3.4	2.5
		MCDB	2.8	2.4	3.0	5.2	10.6	15.0	19.3	17.1	13.6	8.6	4.5	3.8
	10%	WB	0.4	0.0	0.5	2.5	6.2	9.8	13.6	12.4	9.9	6.3	2.3	1.4
		MCDB	1.6	0.9	1.8	3.9	8.5	12.7	17.7	15.6	12.2	7.7	3.5	2.7
Mean Daily Temperature Range	5% DB	MDBR	4.5	4.7	4.3	4.3	4.6	4.9	5.6	5.0	4.1	3.5	3.9	4.3
		MCDBR	4.6	5.0	4.4	5.0	8.0	9.3	9.7	8.4	6.2	4.3	4.1	4.4
		MCWBR	4.3	4.6	3.7	3.6	4.6	5.2	5.1	4.6	3.8	3.3	3.8	4.0
	5% WB	MCDBR	4.6	4.9	4.3	4.8	7.4	8.7	9.3	7.8	5.8	3.9	4.1	4.6
		MCWBR	4.6	4.7	3.9	3.6	4.5	5.1	5.1	4.6	3.8	3.3	4.0	4.3
Clear-Sky Solar Irradiance	taub	N/A	0.258	0.314	0.332	0.345	0.340	0.349	0.335	0.312	0.262	N/A	N/A	
	taud	N/A	2.055	2.154	2.260	2.380	2.483	2.491	2.514	2.514	2.159	N/A	N/A	
	Ebn at Noon	0	413	658	790	833	852	828	796	705	405	0	0	
	Edh at Noon	0	38	75	94	95	89	85	74	55	32	0	0	
All-Sky Solar Radiation	RadAvg	0.00	0.21	1.14	2.91	4.38	4.81	4.34	2.96	1.54	0.42	0.00	0.00	
	RadStd	0.00	0.03	0.13	0.28	0.37	0.44	0.46	0.27	0.17	0.05	0.00	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page